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**3** A balloon contains helium gas.

**(a)** Explain how the particles of helium gas in the balloon exert a pressure on the inside of the balloon.

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..... [3]

**(b)** The balloon contains a volume of 12 $m^3$ of helium gas at a pressure of 110 kPa.

The balloon is released and rises to a height *h* where the pressure of the gas is 52 kPa.

The temperature and the mass of helium gas remain constant.

**(i)** Calculate the volume of the helium gas at height *h*.

volume = ..... $m^3$ [2]

**(ii)** At height *h*, the balloon has gained 120 kJ of energy in the gravitational potential store.

The mass of the balloon is 2.5 kg.

Calculate the height *h*.

*h* = ..... m [3]

[Total: 8]